TUTORIAL 04 · PART II THEORY

THE FACTOR ZOO

因子动物园 — Theory and Taxonomy of Equity Factors. Every subsequent factor tutorial drills into one family; this one builds the asset-pricing lineage, the taxonomy, the "why do factors earn premia" debate, and the sober statistical reality of the factor zoo.



SECTION 1–2

WHAT IS A FACTOR? 什么是因子？

A **factor** in equity research is a systematic, cross-sectionally observable characteristic — or return series constructed from such a characteristic — that explains differences in expected returns across stocks. If we write the return-generating process as:

$$r_{i,t} = E[r_{i,t}] + \sum_{k=1}^{K} \beta_{i,k,t} f_{k,t} + \epsilon_{i,t}$$

then $f_{k,t}$ is a **factor return**, $\beta_{i,k,t}$ is the stock's **loading**, and $\epsilon_{i,t}$ is idiosyncratic noise. A factor is useful if it reduces the covariance matrix to a low-rank-plus-diagonal structure and/or commands a non-zero risk premium

$$\lambda_k = E[f_{k,t}]$$

Critical epistemic point: "Anomaly" is model-relative, not absolute. As benchmark models expand (CAPM → FF3 → Carhart4 → FF5 → FF6), yesterday's anomaly becomes today's risk factor.

CHARACTERISTIC 特征

A firm-level observable at time t — e.g. book-to-market ratio $B/M_{i,t}$

FACTOR 因子

A return series (usually long-short) built by sorting stocks on a characteristic — e.g. HML

ANOMALY 异象

An empirical pattern a benchmark model fails to explain — model-relative, not absolute

ALPHA A 阿尔法

The intercept in a factor-model regression: return unexplained by the chosen factors

Risk factors earn premia as compensation for bad-state exposure; mispricing factors capture the slow correction of behavioral or friction-driven price deviations — most documented factors are argued to be some mixture of both.

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SECTION 2

THEORETICAL LINEAGE 理论谱系

CAPM (1964–66)

Sharpe, Lintner, Mossin. One factor: market beta. The Security Market Line predicts every asset lies on $E[r_i] - r_f = \beta_i(E[r_m] - r_f)$. Empirically rejected: flat SML, size effect, value effect, Roll's critique.

APT (ROSS 1976)

Drops mean-variance machinery; assumes linear factor structure plus no-arbitrage. Justifies "find K common factors" empirically without requiring utility maximization. Expected returns are approximately linear in factor loadings: $E[r_i] \approx r_f + \sum_k \beta_{ik}\lambda_k$.

FF3 (1993)

MKT + SMB (size) + HML (value). 2×3 independent sort: NYSE median size breakpoint; 30th/70th B/M breakpoints. Dramatically improved cross-section fit but could not explain momentum.

CARHART 4 (1997)

FF3 + UMD (12-1 momentum, skipping last month). Rebalanced monthly — reflecting momentum's short signal half-life. Became the standard academic benchmark for a decade.

FF5 (2015) & Q-FACTOR (HXZ 2015)

FF5 adds RMW (profitability) + CMA (investment), motivated by dividend-discount-model decomposition — HML becomes largely redundant. q-factor derives same premia from production-side q-theory: ME, I/A, ROE. Two independent routes converging validates the profitability/investment premium.

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SECTION 3–4

GRS TEST & WHY FACTORS EARN PREMIA 检验与溢价来源

1

TIME-SERIES VS. CROSS-SECTIONAL TESTS

Time-series regressions test whether a portfolio's alpha is zero given factors: $r_{p,t} - r_{f,t} = \alpha_p + \beta_p f_t + \epsilon_{p,t}$. The **GRS test** asks whether all test-asset alphas are jointly zero — preventing one pretty alpha from hiding broad model failure. Sensitive to test-asset choice; one diagnostic, not a final verdict. 防止单一漂亮阿尔法掩盖模型普遍失败。

2

RISK COMPENSATION 风险补偿

Exposure to factor f_k raises the volatility of consumption/wealth in bad states (recessions, liquidity crunches, tail events). The premium $\lambda_k > 0$ is demanded by rational, risk-averse investors — permanent because bearing the risk is unavoidable in equilibrium.

3

BEHAVIORAL MISPRICING 行为错误定价

Prices deviate from fundamental value due to investor psychology (overreaction, underreaction, extrapolation, lottery preference) or market frictions (limits to arbitrage, shorting costs, funding constraints). Premia can erode as more capital chases them or as trading costs rise. 原则上可被套利消除。

No serious practitioner needs a factor to belong purely to one camp. What matters: does the story predict when the factor should fail? Value fails when distress risk is realized; momentum crashes at sharp rebounds; low volatility suffers when leverage constraints relax.

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SECTION 5

THE FACTOR ZOO PROBLEM 因子动物园问题

Scale of the Problem 问题规模

400+

Hundreds of published anomalies cannot all be independent sources of expected return. Some are aliases for the same latent exposure, some are subsumed by known factors, some are too expensive to trade, and some are false discoveries produced by multiple testing on the same finite dataset. 数百个已发表异象不可能都是独立预期收益来源。

The number of factors that "survive against an expanding set of controls" is far smaller than the number "in the literature." A t-stat of 2.0 that was once the publication threshold implies a false-discovery rate of roughly 50% under realistic multiple-testing conditions.

REDUNDANCY 冗余

Many claimed factors are linear combinations of MKT, SMB, HML, UMD — correlated aliases that add no independent pricing information.

DATA MINING 数据挖掘

Thousands of characteristics tried on the same US equity sample. Even random characteristics pass a t>2 filter with high frequency given enough tries.

MULTIPLE-TESTING DISCIPLINE 多重检验纪律

Harvey, Liu & Zhu (2016): adjust thresholds upward (t > 3.0 for new factors). Benjamini-Hochberg FDR control. Out-of-sample and international replication are the gold standards.

MINIMUM STANDARD 最低标准

Economic rationale · point-in-time construction · robustness across periods and universes · low correlation to existing factors · implementable net of costs · multiple-testing-aware validation.

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SECTIONS 6–7

FACTOR TAXONOMY & SCORECARD 因子分类与评分卡

MASTER TAXONOMY 分类总表

Family 族	Representative Signals 代表信号
Value 价值	B/P, E/P, EBIT/EV, CFO/P
Momentum 动量	12-1 return, residual mom, earnings mom
Quality 质量	GP/A, ROE, accruals, F-score, QMJ
Investment 投资	Asset growth, capex growth, net issuance
Low Risk 低风险	Beta, idio-vol, downside vol
Liquidity 流动性	Amihud, turnover, spread, ADV
Sentiment 情绪	Analyst revisions, short interest, news tone
Alt Data 另类数据	Web traffic, app usage, supply chain text

RISK VS. MISPRICING SCORECARD 逐因子评分

Factor	Risk Story	Behavioral Story
Value	Distress & duration risk	Extrapolation, overreaction
Momentum	Crash & optionality exposure	Underreaction, slow diffusion
Quality	Defensive cash flows	Underappreciated persistence
Investment	q-theory investment risk	Overinvestment, empire building
Low Vol	Leverage constraints (BAB)	Lottery demand for high beta
Size	Illiquidity / funding risk	Neglected firms

The scorecard is not a verdict — it is a prompt. For every factor, write down what evidence would convince you it is not real.

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SECTIONS 8–9

CONSTRUCTION STYLES & FACTOR TIMING 构建方式与因子择时

LONG-SHORT ACADEMIC FACTOR

Decile or 2×3 sort; value-weighted; tests whether the premium exists in principle. Reports pure factor return. Not directly investable due to shorting costs, turnover, and capacity.

RANK-WEIGHTED DOLLAR-NEUTRAL

Weights proportional to cross-sectional rank; dollar-neutral by construction; smoother exposure than decile sorts; still requires shorting but reduces turnover concentration.

LONG-ONLY BENCHMARK TILT

Factor expressed as overweights/underweights vs. a benchmark. Tests benchmark-relative stock selection under real constraints. Always report: weighting, rebalancing frequency, breakpoint universe, sector/beta neutrality.

FACTOR TIMING & CROWDING 择时与拥挤

Valuation spreads, macro regimes, crowding indicators can contain information — but add another layer of multiple testing. Safer default: diversification across factors with drawdown-aware risk budgeting. Monitor common-manager holdings, ETF flows, short interest, borrow cost, turnover spikes.

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SECTIONS 10–11

US VS A-SHARES & THE 2×3 SORT 美股与 A 股 · 算例

FACTORS IN US VS A-SHARES 市场差异

MARKET STRUCTURE 市场结构

A-share shorting constraints are more severe; limit-up/limit-down events distort momentum signals; disclosure quality and earnings-management incentives differ significantly from US markets.

MOMENTUM 动量

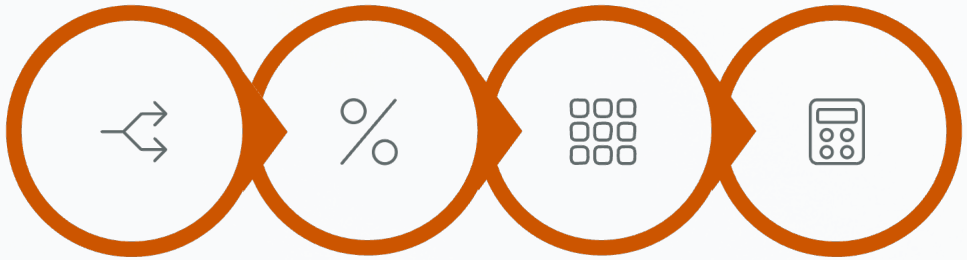
US momentum is a robust canonical factor. A-share momentum is more horizon- and limit-event-dependent — treat any cross-market factor import as a hypothesis, not a parameter-free truth.

SIZE & QUALITY A 股规模 / 质量

A-share size has been shaped by shell value and quant crowding. Quality screens must be more alert to subsidies, related-party activity, and earnings-management incentives unique to the A-share regulatory regime.

WORKED EXAMPLE: THE 2×3 SORT 算例

A Fama-French-style 2×3 sort on a toy universe works as follows:



Independent sorting keeps the two dimensions near-orthogonal by construction. The same template applies to profitability, investment, and custom factors. Always use NYSE breakpoints (not the full universe) to avoid microcap domination of the sort breakpoints.

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FROM FACTORS TO STRATEGIES 从因子到策略

1 ECONOMIC RATIONALE & SIGNAL 经济逻辑与信号

Steps 1–2

- Articulate the theoretical reason before seeing the backtest
- Build a clean, point-in-time signal with no look-ahead
- Confirm IC, sorts, and regressions agree directionally

2 SINGLE-FACTOR & INCREMENTALITY TESTS 单因子与增量检验

Steps 3–4

- Single-factor sort across reasonable horizons and universes
- Incrementality vs. known factors (FF5, q5, Carhart)
- Apply multiple-testing-aware thresholds (t > 3.0)

3 BACKTEST & RED TEAM 回测与红队

Steps 5–6

- Walk-forward or out-of-sample evaluation
- Red-team stress: different universes, sub-periods, geographies
- Cost and turnover realism — borrow cost, market impact

4 CONSTRAINED PORTFOLIO → LIVE 约束组合到实盘

Steps 7–9

- Constrained portfolio construction with universe rules and risk limits
- Paper trade with full execution assumptions and capacity estimates
- Small live allocation with monitoring and failure-scenario governance

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SECTIONS 13–14

PRACTICAL CHECKLIST & SUMMARY 实践清单与小结

WHAT MAKES A GOOD FACTOR 好因子的标准

- It is known at the time it is used — no look-ahead, point-in-time construction 使用时确实已知
- It has a reason to exist before seeing the backtest — theory first 在看到回测前就有存在理由
- It survives neutralization to obvious exposures (sector, size, beta) 对明显暴露中性化后仍存在
- IC, portfolio sorts, and regressions agree directionally IC 、分层和回归方向一致
- It works across reasonable horizons and universes — not just US large-cap 1990–2020 跨期限和股票池稳健
- It is not just a renamed existing factor — passes incrementality tests 不是已有因子的改名
- It survives costs, turnover, and capacity — t-stat net of friction 能扛住成本和换手
- It has a failure story and a monitoring plan 有失效故事和监控方案

MODEL LINEAGE SUMMARY 模型谱系汇总

Model	Year	Factors #	Grounding
CAPM	1964	1	Mean-variance equilibrium
APT	1976	K	No-arbitrage, linear factors
FF3	1993	3	Empirical + risk rationalization
Carhart 4	1997	4	Empirical momentum add-on
FF5	2015	5	Dividend-discount decomposition
q (HXZ)	2015	4	Investment-based q-theory
q5	2021	5	q-theory + expected growth
Stambaugh-Yuan	2017	4	Behavioral mispricing clusters

The factor zoo is both a library and a warning. Good factor research combines theory, data discipline, conservative testing, cost realism, and humility about regimes. 因子动物园既是资料库，也是警告。

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